www.tntextbooks.in

Points to Remember

- Iteration repeats the two steps of evaluating a condition and executing a statement, as long as the condition is true.
- An expression involving variables, which remains unchanged by an assignment to one of these variables, is called an invariant of the assignment.
- An invariant for the loop body is known as a loop invariant.
- A loop invariant is true at
- (a) at the start of the loop (just before the loop)
- (b) at the start of each iteration (before loop body)
- (c) at the end of each iteration (after loop body)
- (d) at the end of the loop (just after the loop)
- When a loop ends, the loop invariant is true. In addition, the termination condition is also true.
- Recursion must have at least one base case.
- Recursion step breaks the problem into sub-problems of smaller size, assumes solutions for sub-problems are given by recursive calls, and constructs solution to the given problem.
- In recursion, the size of input to a sub-problem must be strictly smaller the size of the given input.

Evaluation

C4VWMI

Part I

1. A loop invariant need not be true

   (a) at the start of the loop. (b) at the start of each iteration

   (c) at the end of each iteration (d) at the start of the algorithm

2. We wish to cover a chessboard with dominoes, the number of black squares and the number of white squares covered by dominoes, respectively, placing a domino can be modeled by

   (a) b := b + 2 (b) w := w + 2 (c) b, w := b+1, w+1 (d) b := w